ECE 210 Midterm 3 Worksheet

Note: This worksheet is not guaranteed to be entirely representative of the midterm's contents. Material may appear on this worksheet which will not appear on the midterm, and vice versa.

Partial Fraction Expansion

1) (Technique 1)

a) sub problem 1

Solution sub problem 1 solution

$$sF(s) = s\frac{s+3}{s(s+2)(s+5)} = ...$$

b) sub problem 2

Solution sub problem 2 solution

2) (Technique 2)

a) sub problem 1

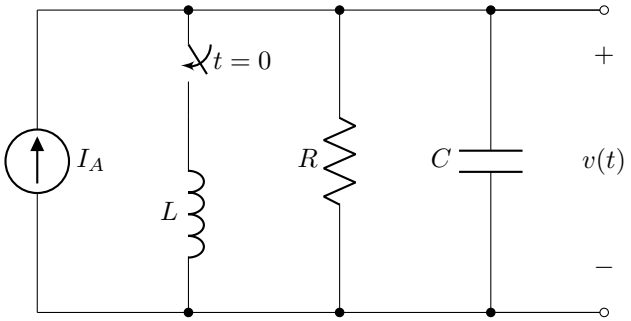
Solution sub problem 1 solution

b) sub problem 2

Solution sub problem 2 solution

S-Domain Circuit Analysis

3) In the circuit below, the switch has been open for a long time and is closed at $t = 0$. Let $I_A = 1$ mA, $L = 2$ H, $R = 1.5$ kΩ, and $C = \frac{1}{6}$ μF.



a) Find the initial conditions $i_L(0^-)$ and $v(0^-)$, and draw the s-domain equivalent circuit for $t \geq 0$.

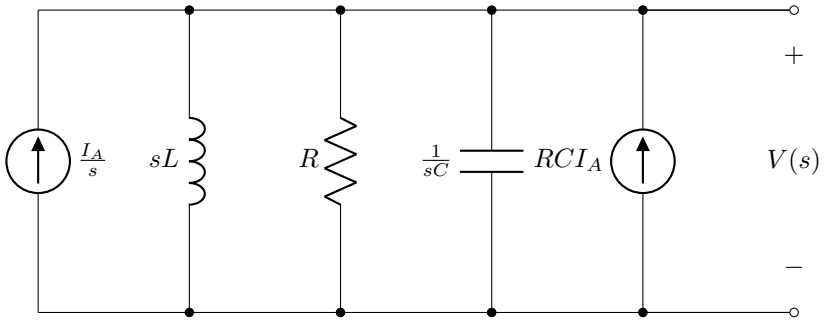
Solution With the switch open for a long time, the circuit is in DC steady state. The inductor branch is disconnected, so $i_L(0^-) = 0$. The capacitor is fully charged, and acts as an open circuit, so all of I_A flows through R:

$$v(0^-) = RI_A = (1.5\ \text{k}\Omega)(1\ \text{mA}) = 1.5\ \text{V}.$$

For $t \geq 0$, the switch is closed, and we redraw the circuit in the s-domain using element transformations. Each energy-storage element becomes its impedance plus a source that accounts for its initial condition. Since all the elements share the same two nodes, we use the parallel (Norton) form for each:

- **Current source:** $i_A(t) = I_A$ for $t \geq 0$, so it becomes $\dfrac{I_A}{s}$.
- **Resistor:** it is unchanged and stays R.
- **Inductor:** from $v_L(t) = L\,\dfrac{di_L}{dt}$ we get $V_L(s) = sL\,I_L(s) - L\,i_L(0)$, which gives $I_L(s) = \dfrac{V_L(s)}{sL} + \dfrac{i_L(0)}{s}$. This is an impedance sL in parallel with a current source $\dfrac{i_L(0)}{s}$. Because $i_L(0^-) = 0$, this source is zero, and the inductor is just sL.
- **Capacitor:** from $i_C(t) = C\,\dfrac{dv_C}{dt}$ we get $I_C(s) = sC\,V_C(s) - C\,v_C(0)$. This is an impedance $\dfrac{1}{sC}$ in parallel with a current source $C\,v_C(0)$. Here $C\,v(0^-) = C\,(RI_A) = RCI_A$.

Putting these together, the s-domain circuit has I_A/s, sL, R, $1/sC$, and the capacitor's RCI_A source all in parallel, with $V(s)$ across the output terminals. This is the circuit drawn below.



s-domain equivalent circuit for $t \geq 0$

b) Find the zero-state and zero-input components of $V(s)$, and then find $v_{zs}(t)$ and $v_{zi}(t)$ for $t \geq 0$.

Solution The equivalent impedance seen by both sources is the parallel combination of sL, R, and $1/sC$:

$$Z_{eq}(s) = \left(\frac{1}{sL} + \frac{1}{R} + sC\right)^{-1} = \frac{sLR}{s^2LRC + sL + R} = \frac{s/C}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}.$$

Zero-state (only the I_A/s source on):

$$V_{zs}(s) = Z_{eq}(s)\,\frac{I_A}{s} = \frac{I_A/C}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}.$$

Zero-input (only the RCI_A source on):

$$V_{zi}(s) = Z_{eq}(s)\,RCI_A = \frac{RI_A\,s}{s^2 + \frac{1}{RC}s + \frac{1}{LC}}.$$

Substitute values. Here $\frac{1}{RC} = 4000$, $\frac{1}{LC} = 3 \times 10^6$, $\frac{I_A}{C} = 6000$, and $RI_A = 1.5$. The denominator factors as

$$s^2 + 4000s + 3 \times 10^6 = (s + 1000)(s + 3000).$$

Zero-state, by partial fractions:

$$V_{zs}(s) = \frac{6000}{(s + 1000)(s + 3000)} = \frac{3}{s + 1000} + \frac{-3}{s + 3000}$$

$$\boxed{v_{zs}(t) = \left[3e^{-1000t} - 3e^{-3000t}\right] u(t)\ \text{V}}$$

Zero-input, by partial fractions:

$$V_{zi}(s) = \frac{1.5\,s}{(s + 1000)(s + 3000)} = \frac{-0.75}{s + 1000} + \frac{2.25}{s + 3000}$$

$$\boxed{v_{zi}(t) = \left[-0.75e^{-1000t} + 2.25e^{-3000t}\right] u(t)\ \text{V}}$$

Marginal Stability & Resonance

4) Consider the causal linear time-invariant system

$$\frac{d^2y(t)}{dt^2} + \omega_0^2 y(t) = \omega_0^2 x(t), \qquad \omega_0 = 2 \text{ rad/s}.$$

The input and output are dimensionless, and t is measured in seconds. Use zero initial conditions in parts (b) and (c).

a) Find $H(s)$ and its poles. Find the zero-input response for arbitrary finite $y(0) = y_0$ and $y'(0) = v_0$. Is the natural response asymptotically stable, marginally stable, or unstable? Is the system bounded-input, bounded-output (BIBO) stable? Justify both answers.

Solution With zero initial conditions,

$$H(s) = \frac{\omega_0^2}{s^2 + \omega_0^2}, \qquad s = \pm j\omega_0 = \pm j2 \text{ s}^{-1}.$$

The zero-input response is

$$y_{zi}(t) = y_0 \cos(\omega_0 t) + \frac{v_0}{\omega_0} \sin(\omega_0 t), \qquad t \geq 0.$$

Every finite initial state produces a bounded natural response, but nonzero oscillations do not decay. Thus the natural response is **marginally stable**, not asymptotically stable: the imaginary-axis poles are simple. However, the system is **not BIBO stable**. Its impulse response $h(t) = \omega_0 \sin(\omega_0 t)u(t)$ is not absolutely integrable. Part (c) also supplies a bounded input with an unbounded output.

b) Find the zero-state output for $x(t) = \cos(\Omega t)u(t)$, where $\Omega = 1$ rad/s. Is the output bounded? Does the natural-frequency component decay?

Solution Using $X(s) = s/(s^2 + \Omega^2)$ and $\Omega \neq \omega_0$,

$$Y(s) = \frac{\omega_0^2 s}{(s^2 + \omega_0^2)(s^2 + \Omega^2)} = \frac{\omega_0^2}{\omega_0^2 - \Omega^2} \left(\frac{s}{s^2 + \Omega^2} - \frac{s}{s^2 + \omega_0^2} \right).$$

Therefore,

$$y(t) = \frac{\omega_0^2}{\omega_0^2 - \Omega^2} [\cos(\Omega t) - \cos(\omega_0 t)]u(t) = \frac{4}{3} [\cos(\Omega t) - \cos(\omega_0 t)]u(t).$$

The output is bounded ($|y(t)| \leq 8/3$). The natural-frequency term persists: there is no damping. A bounded response to this particular input does not imply BIBO stability.

c) Repeat part (b) for $\Omega = \omega_0$. Explain resonance using the poles of $Y(s)$, and explain why the result is consistent with part (a).

Solution At resonance, the input frequency matches the natural frequency:

$$Y(s) = \frac{\omega_0^2 s}{(s^2 + \omega_0^2)^2}, \qquad y(t) = \frac{\omega_0 t}{2} \sin(\omega_0 t)u(t).$$

Using $\mathcal{L}\{tf(t)\} = -dF(s)/ds$, we obtain

$$\mathcal{L}\{t \sin(\omega_0 t)u(t)\} = \frac{2\omega_0 s}{(s^2 + \omega_0^2)^2}.$$

The poles of $Y(s)$ at $\pm j\omega_0$ are now **double**, producing a linearly growing oscillation envelope $\omega_0 t/2$. The input is bounded, but the output is unbounded. This is resonance and proves the failure of BIBO stability. It does not contradict marginal stability, which describes the zero-input natural response; the poles of $H(s)$ remain simple.

5) problem 2

a) marginal stability part

Solution sub problem 1 solution

b) resonance part

Solution sub problem 2 solution

c) etc

Solution sub problem 3 solution

Fourier Transforms

6) Find the Fourier transform:

$$g(t) = \frac{6}{t^2 + 9}$$

Solution This problem looks like the Fourier transform pair

$$f(t) = e^{-a|t|} \longleftrightarrow F(\omega) = \frac{2a}{a^2 + \omega^2}$$

with $a = 3$. However, we have $f(t)$, not $F(\omega)$ matching the right-hand side.

To take care of this, we can use the Symmetry Property, which states that if $f(t) \longleftrightarrow F(\omega)$, then

$$F(t) \longleftrightarrow 2\pi f(-\omega)$$

Now, with $a = 3$, we can evaluate: $f(-w) = e^{-3|-\omega|} = e^{-3|\omega|}$. Thus with the Symmetry Property,

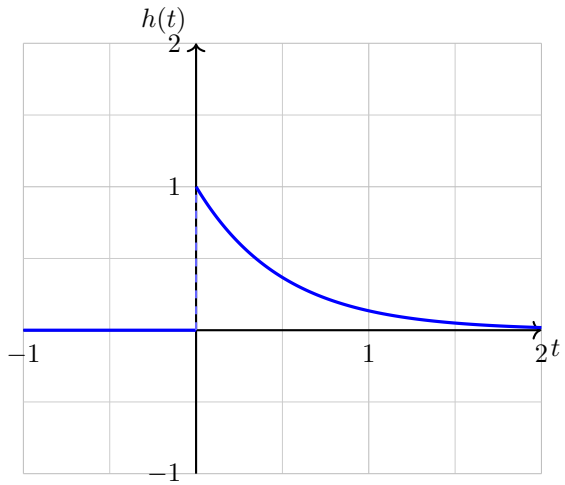
$$F(t) = \frac{2(3)}{3^2 + t^2} = \frac{6}{t^2 + 9} \longleftrightarrow 2\pi f(-\omega) = 2\pi e^{-3|\omega|}$$

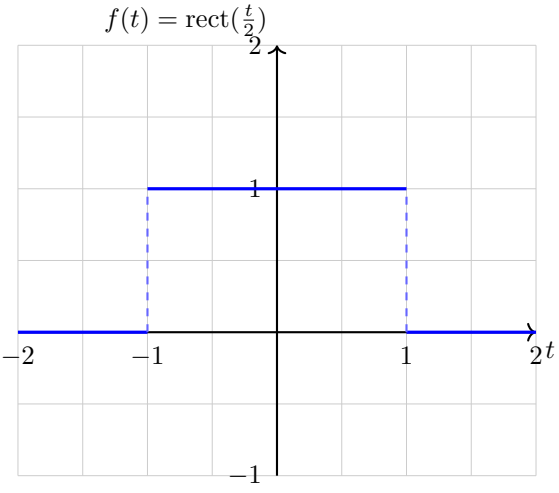
Finally, since $g(t) = F(t)$, we can conclude that the Fourier transform is $G(\omega) = 2\pi e^{-|\omega|}$.

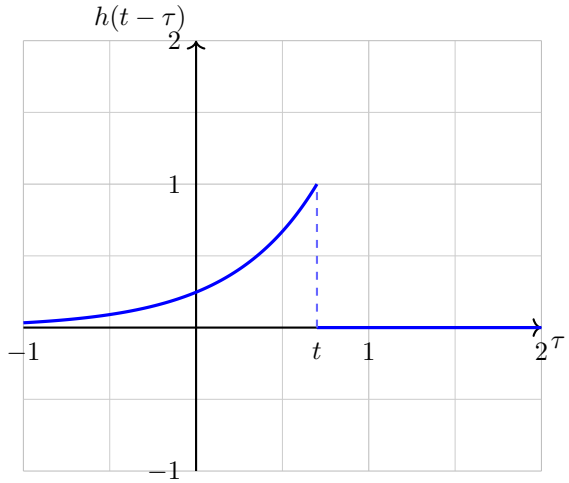
7 Find the output:

$$y(t) = e^{-2t}u(t) * \text{rect}(\frac{t}{2})$$

Solution Let $h(t) = e^{-2t}u(t)$ and $f(t) = \text{rect}(\frac{t}{2})$







$$y(t) = \int_{-\infty}^{\infty} h(t-\tau)f(\tau)\mathrm{d}\tau$$

For $t < -1$: $y(t) = 0$ For $-1 \le t < 1$:

$$\begin{aligned} y(t) &= \int_{-1}^{t} e^{-2(t-\tau)}\mathrm{d}\tau = e^{-2t}\int_{-1}^{t} e^{2\tau}\mathrm{d}\tau \\ &= e^{-2t}\cdot\left[\frac{1}{2}e^{2\tau}\right]_{-1}^{t} = e^{-2t}\left[\frac{1}{2}e^{2t} - \frac{1}{2}e^{-2}\right] = \frac{1}{2}\left[1 - e^{-2(t+1)}\right] \end{aligned}$$

For $t \ge 1$:

$$\begin{aligned} y(t) &= \int_{-1}^{1} e^{-2(t-\tau)}\mathrm{d}\tau = e^{-2t}\int_{-1}^{1} e^{2\tau}\mathrm{d}\tau \\ &= e^{-2t}\cdot\left[\frac{1}{2}e^{2\tau}\right]_{-1}^{1} = e^{-2t}\left[\frac{1}{2}e^{2} - \frac{1}{2}e^{-2}\right] = \frac{1}{2}(e^{2} - e^{-2})e^{-2t} \end{aligned}$$

So,

$$\boxed{y(t) = \begin{cases} 0, & \text{if } t < -1 \\ \frac{1}{2}\left[1 - e^{-2(t+1)}\right], & \text{if } -1 \le t < 1 \\ \frac{1}{2}(e^2 - e^{-2})e^{-2t}, & \text{if } t \ge 1 \end{cases}}$$